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PALEOMAGNETISM OF THE GAYS RIVER ZINC-LEAD DEPOSIT, NOVA SCOTIA: PENNSYLVANIAN ORE GENESIS

H. Pan

Dept. of Geophysics, Univ. of Western Ontario, London, Canada, N6A 5B7

D.T.A. Symons

Dept. of Geology, Univ. of Windsor, Windsor, Canada, N9B 3P4

D.F. Sangster

Geological Survey of Canada, Ottawa, Canada, K1A 0E8

Abstract. The Gays River zinc-lead deposit is the largest known Mississippi Valley-type deposit in the Mississippian Windsor Group of Atlantic Canada. Thermal and AF demagnetization show that both ore and host rock have very similar magnetic characteristics. Stability tests constrain the magnetization to be mid-Mississippian to late Pennsylvanian. The pole position from the ore and host rock sites falls next to the Pennsylvanian APW path, suggesting an 8° clockwise rotation of the Meguma platform relative to the craton. Regional thermal history and geochemical evidence suggest that the zinc-lead ore magnetization and host rock remagnetization are coeval.

Introduction

The Gays River deposit (GRD) is the largest Mississippi Valley-type (MVT) zinc-lead deposit in the Mississippian Windsor Group of Atlantic Canada with an estimated reserve of 2.4×10^6 t. Unlike other MVT deposits in North America its homogenization temperatures of sphalerite fluid inclusions range to >210° C [Akande and Zentilli, 1984; Ravenhurst et al., 1989; Kontak, 1992]. Also, the GRD mineralization was formed mainly by infilling of primary porosity rather than collapse structures. The Mississippian host rocks rule out a genetic link with other Appalachian MVT deposits which appear to be formed during the Devonian Acadian orogeny [Nakai et al., 1990; Kesler and Van der Pluijm, 1990; Pan and Symons, 1993]. Attempts to date the GRD mineralization using zircon and apatite fission track [Ravenhurst et al., 1989; Arne et al., 1990] and $^{40}Ar/^{39}Ar$ and K-Ar isotopic [Ravenhurst et al., 1989] techniques have been inconclusive. In this study, we used paleomagnetic methods to date its genesis.

Geology

Basement rocks at Gays River (Fig. 1), between the Schubenacadie and Musquodoboit basins, consist of folded greenschist facies metasediments of the Cambro-Ordovician Meguma Group into which Devonian granitoid plutons were emplaced during the Acadian orogeny [Keppie, 1982]. Early Mississippian red beds of the Horton Group were deposited on this basement, and then carbonates, evaporites and siltstones of the Visean (333-352 Ma) Windsor Group. Alteration and mineralization of dolostones in the Gays River Formation of the Windsor Group took place without a volume change, preserving primary porosity and fossil structures [Schenk, 1984]. A basal breccia of a near-sedimentary karstic origin has fragments up to several meters across [Schenk, 1984]. The GRD is unconformably overlain by unconsolidated Lower Cretaceous sediments [Kontak, 1992], placing a younger age limit on the mineralization [Davies et al., 1984]. It consists of sphalerite and galena with trace amounts of Fe, Ba, Cu, Ag, and Cd. It ranges from massive (sulphides > 80%) to disseminated (<10%). A well-defined paragenetic mineral sequence shows the GRD to be epigenetic [Akande, 1982; Kontak, 1992]. The orebody is stratabound but contains vein-like replacements of solution features in host dolostones. Both forms exhibit the same geochemical features, indicating a common mineralization event [Akande, 1982; Kontak, 1992]. Homogenization temperatures from fluid inclusions in sphalerite range from 90° to >210° C [Kontak, 1992] to set the minimum temperature for ore formation.

Sampling and Experimental Methods

Four to six oriented block samples were collected at 21 sites at the GRD and at 6 sites in limestones/dolostones at the same

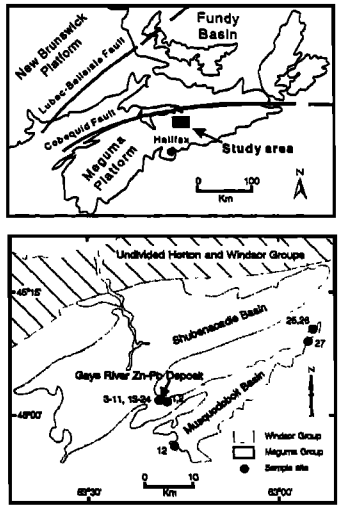


Fig. 1. Geology and site locations in the Gays River area.

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stratigraphic level elsewhere (Fig. 1). At the mine, site 22 is from unmineralized evaporite just above the orebody and sites 9 and 15 are from the unmineralized basal breccia. Ore samples range from disseminated to massive sphalerite with galena. Three or four specimens were drilled from most blocks to give 421 specimens. They were stored and processed in a shielded room. Their natural remanent magnetization (NRM) was measured using a CTF magnetometer with a noise level of $<4\times10^{-6}$ A/m. All specimens were step demagnetized using MMTD1 or TSD-1 thermal units or a Sapphire Instruments SI-3 alternating field (AF) unit. Characteristic remanent magnetization (ChRM) components were obtained using the Kirschvink [1980] method, requiring the maximum angular deviation to be <10°. Site and unit mean directions were calculated following Fisher [1953].

Results

Most NRM directions are southward with shallow positive inclinations. Almost all specimens show a two-component magnetization on step demagnetization including low to high-grade ore, dolostone, and breccia (Fig. 2). A viscous remanence close to the present Earth's magnetic field direction is first removed up to 200°- 300°C or to 30-40 mT. Thereafter the ChRM component decays linearly to the origin and is mostly demagnetized at 500°C or 130 mT (Fig. 2). Three sites (#25, 26, 27) were sampled in dipping Gays River dolostones, but two (#25, 26) gave incoherent directions, negating the paleomagnetic fold test. The evaporite NRM intensities (site 22) were too weak to measure. The breccia clasts (sites 9 and 15) have similar magnetic properties and ChRM direction as the ore and dolostone so that both breccia tests are negative, indicating that the ChRM is postbrecciation (Fig. 3b and 3c). Isothermal remanent magnetization (IRM) acquisition curves for 10 specimens from various lithologies show >90% saturation by 300 mT and AF demagnetization of the saturation IRM indicate that single to pseudosingle domain magnetite is the sole magnetic carrier. The site mean directions for ore, dolostone and breccia clasts are the same at >95% confidence, indicating a coeval magnetization. Their combined unit mean direction is Dec = 177.6°, Inc = 14.4° ($k = 40.7$, $\alpha_{95} = 4.9°$, N = 22) (Fig. 3a) to give a pole of 37.6°N, 120.0°E (dp = 2.6°, dm = 5.0°) (Fig. 4, Table 1).

Timing of Acquisition of Characteristic Magnetization

Ore samples with >90% of sphalerite have virtually the same ChRM characteristics and directions as lower grade ore, dolostone and breccia units. Thus the magnetization must be coeval with or later than mineralization. Combined with the two negative breccia tests and epigenetic paragenetic sequence, the magnetization must postdate the mid-Mississippian host rocks.

Scotese et al. [1984] reported a positive paleomagnetic fold test on rocks from several horizons nearby in the Windsor Group, indicating a pre-Alleghenian orogeny or pre-late Pennsylvanian magnetization. Our fold test is inconclusive, but our samples show similar ChRM characteristics to those found by Scotese et al. [1984] except for a 14° difference in declination. Most of our sites are near the mine whereas those of Scotese et al. [1984] are much further away, so that the discordance could be caused by rotations on known postore faults near the mine. The apparent polar wander (APW) path for Atlantic

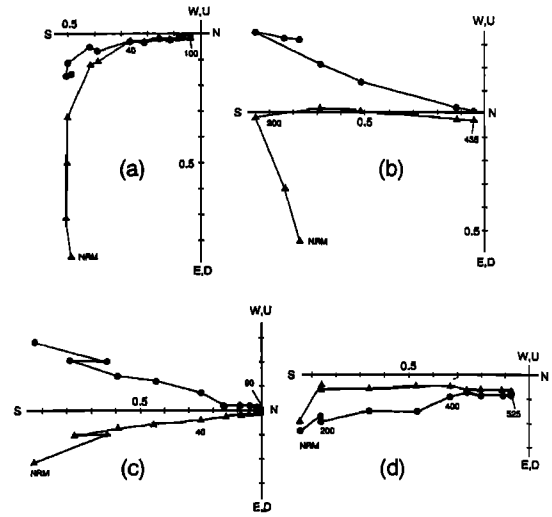


Fig. 2 Orthogonal demagnetization diagram of representative specimens showing the two-component magnetization revealed by AF and thermal demagnetization in: (a) and (b) dolostone; and (c) and (d) ore. Circles and triangles are projections on the horizontal and vertical planes, respectively. Axis units are normalized to the initial NRM intensity. The step intensities are in mT and °C for AF (a, c) and thermal (b, d) demagnetization.

Canada from Pennsylvanian to Triassic is defined mostly by increasing inclination. The fact that both studies obtain the same ChRM inclination indicates that the host rock and ore acquired their magnetization at the same time, requiring dolomitization and remagnetization before folding.

The GRD pole position falls beside the early Pennsylvanian (300-320 Ma) segment of the North American APW path [Van der Voo, 1990] (Fig. 4), fitting in the late Mississippian to mid-Pennsylvanian (300-330 Ma) window permitted by the stability tests. The offset of the GRD pole can be eliminated by rotating the Meguma platform anticlockwise by 8° relative to the craton

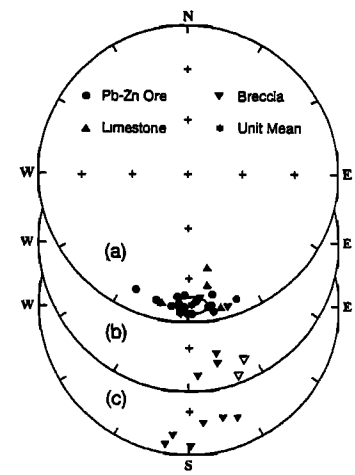


Fig. 3. (a) Equal-area stereogram of the site mean ChRM directions for various lithologies. The unit mean direction is circumscribed by its cone of 95% confidence. (b) and (c) block mean ChRM directions of basal breccia clasts from sites 15 and 9, respectively. Solid symbols represent the projections on the lower hemisphere and open symbols on the upper hemisphere.

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TABLE 1. Site Mean Results

Site§	Lith.	Spec. N/N_o	Dec./Inc. (°)	k	α_{95} (°)
1	D	12/13	169/27	92	5
2	D	13/14	168/36	31	8
3	Om	16/16	187/13	27	7
4	Od	09/16	204/17	39	8
5	Om	15/16	182/20	56	5
6	Om	14/16	179/07	36	7
7	Od	12/14	194/14	82	5
8	Om	14/16	180/07	28	8
9	B	6+/13	175/17	30	10
10	Ol	15/16	185/19	43	6
11	Ol	05/16	182/11	29	15
12	D	13/13	192/13	46	6
13	Oh	16/16	169/07	60	5
14	Oh	14/16	168/19	25	8
15	B	5+/10	164/08	14	18
16	Oh	14/16	159/11	47	6
17	Ol	08/16	171/11	92	6
18	Ol	05/16	170/12	73	9
19	Om	07/16	177/15	128	5
20	Oh	14/16	183/09	22	9
23	Ol	04/16	184/13	127	8
27	D	10/14	166/08	49	7
Unit Mean:		Dec. = 177.6°; Inc. = 14.4°, N = 22, k = 40.7, α_{95} = 4.9°			
Pole Position:		37.6°N, 120.0°E, dp = 2.6°, dm = 5.0°			

Notes. Lithology (Lith.): D, dolostone; B, breccia; E, evaporite; Ol, Om, Oh, low (<10% ZnS+PbS), medium (10-80%) and high (>80%) grade ore. N/N_o are the number of specimens measured and used to compute the site mean direction. Dec., Inc., k, α_{95}, are the declination, inclination, precision parameter and radius of 95% confidence of the mean, and dp, dm the radii of 95% confidence of the pole, respectively (Fisher, 1953). + number of clasts. § sites 21 (Om), 22(E), 24(Ol), 25(D), and 26(D) were incoherent.

(Fig. 4), possibly on the Cobequid Fault as Symons et al. [1989] argued from paleomagnetic results for the Triassic Fundy Group.

Relationship between Magnetization and Mineralization

Radiometric evidence suggests that mineralization is coeval with magnetization within a few million years. $^{40}Ar/^{39}Ar$ ages from Devonian rocks southwest of GRD indicate that a pervasive thermal event reset the argon clock in micas and K-feldspars at 300-320 Ma [Reynolds et al., 1981, 1987; Muecke et al., 1988]. Just south of the GRD, Reynolds et al. [1981] found a $^{40}Ar/^{39}Ar$ age of 300 Ma for muscovite from an epithermal Pb-Ag deposit in the Musquodoboit batholith. Near the GRD a Rb-Sr isochron age of 300 ± 6 Ma on illite from the Horton-Windsor disconformity records a strontium homogenization event associated with major hydrothermal fluid flow [Ravenhurst et al., 1990]. The GRD's Pennsylvanian magnetization fits with this hydrothermal mineralization event. Also fission track data from detrital apatites in the Meguma Group

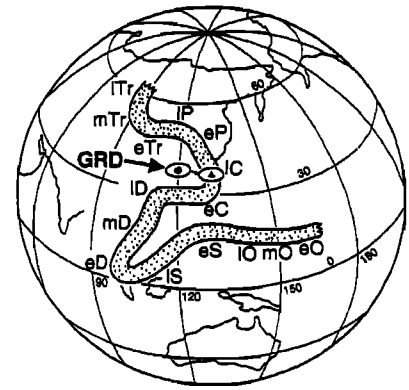


Fig. 4. Gays River deposit (GRD) pole position with its oval of 95% confidence before (circle) and after (triangle) 8° anticlockwise rotation of the Meguma platform relative to the craton. Apparent polar wander path from Early Ordovician (eO) to Late Triassic (lTr) for North America is from Van der Voo [1990].

near GRD suggest that sulphide mineralization must predate 220-240 Ma [Arne et al., 1990].

Fluid inclusion and carbon and oxygen isotope studies show that the mineralizing fluids were >210° C. However the sedimentary cover was <900 m, too thin to explain the high temperatures assuming a normal geothermal gradient [Boehner, 1981]. Thus a pervasive hydrothermal event is needed to sustain the high geothermal gradients required for mineralization [Akande and Zentilli, 1984; Ravenhurst et al., 1989; Arne et al., 1990; Kontak, 1992].

The chemical conditions are still unknown for precipitating sulphide ores with magnetite or for resetting the host rocks' magnetization. Recent studies on hydrocarbon-impregnated sediments have shown their remagnetization to be related to biodegradation of hydrocarbon causing magnetite precipitation [McCabe et al., 1987; Elmore and Crawford, 1988; Elmore and Leach, 1990]. GRD is atop a carbonate mound with the form of a classic hydrocarbon trap. Akande and Zentilli [1984] suggested that zinc-lead geopetal precipitation reflects hydrocarbon saturation during mineralization. Ravenhurst et al. [1989] noted pyrobitumen throughout the GRD and methane in fluid inclusions in ore minerals. Thus hydrocarbons were available to precipitate trace magnetite with the sphalerite and galena and to reset the host rocks' magnetization.

Conclusions

Both ore and host rocks at GRD carry the same ChRM. Paleomagnetic stability tests constrain the magnetization to be post-mid-Mississippian to pre-late Pennsylvanian. The GRD pole agrees, falling beside the mid-Pennsylvanian segment of the APW path. Radiometric dating, high mineralization temperatures, and geochemical evidence relate the hydrothermal mineralization and magnetization events. The small offset of the GRD pole from the APW path is explained by an 8° clockwise rotation of the Meguma platform relative to the craton, consistent with a previous paleomagnetic study.

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References

Akande, S.O., Genesis of the lead and zinc mineralization at Gays River, Nova Scotia, Canada: A geologic, fluid inclusion and stable isotopic study, Ph.D. thesis, Dalhousie Univ., 349 p., 1982.

Akande, S.O., and Zentilli, M., Geologic, fluid inclusion, and stable isotope studies of the Gays River lead-zinc deposit, Nova Scotia, Canada, Econ. Geol., 79, 1187-1211, 1984.

Arne, D.C., Duddy, I.R., and Sangster, D.F., Thermochronologic constraints on ore formation at the Gays River Pb-Zn deposit, Nova Scotia, Canada, from apatite fission track analysis, Can. J. Earth Sci., 27, 1013-1022, 1990.

Boehner, R.C., Stratigraphy and depositional history of marine evaporite in the lower Carboniferous Windsor Group, Shubenacadie and Musquodoboit structural basins, Nova Scotia, Canada, Nova Scotia Dept. Mines, Open File Rept. 468, pp. 28, 1981.

Davies, E.H., Akande, S.E., and Zentilli, M., Early Cretaceous deposits in the Gays River lead-zinc mine, Nova Scotia, Geol. Surv. Can., Pap. 84-1A, 353-358, 1984.

Elmore, R.D., and Crawford, L., Remanence in authigenic magnetite: Testing the hydrocarbon-magnetite hypothesis, J. Geophys. Res., 95, 4539-4549, 1988.

Elmore, R.D., and Leach, M.C., Remagnetization of the Rush Springs Formation, Cement, Oklahoma: Implications for dating hydrocarbon migration and aeromagnetic exploration, Geology, 18, 124-127, 1990.

Fisher, R.A., Dispersion on a sphere, Proc. R. Soc. London, Ser. A 217, 295-305, 1953.

Keppie, J.D., The Minas geofracture, Geol. Assoc. Can. Spec. Pap. 24, 263-280, 1982.

Kesler, E.S., and van der Pluijm, B., Timing of Mississippi Valley-type mineralization: Relation to Appalachian orogenic events, Geology, 18, 1115-1118, 1990.

Kirschvink, J.L., The least squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc., 62, 699-718, 1980.

Kontak, D.J., A preliminary report on geological, geochemical, fluid inclusion and isotopic studies of the Gays River Zn-Pb deposit, Nova Scotia, Nova Scotia Dept. Nat. Resour., Open File Rep. 92-014, 223 pp., 1992.

McCabe, C., Sassen, R., and Saffer, B., Occurrence of secondary magnetite within biodegraded crude oil, Geology, 15, 7-10, 1987.

Muecke, G.K., Elias, P., and Reynolds, P.H., Hercynian/Alleghenian overprinting of an Acadian terrain: $^{40}Ar/^{39}Ar$ studies in the Meguma Zone, Nova Scotia, Canada, Chemical Geol. (Isotope Geoscience Section), 73, 153-167, 1988.

Nakai, S., Halliday, A.N., Kesler, S.E., and Jones, H.D., Rb-Sr dating of sphalerites from Tennessee and the genesis of Mississippi Valley type ore deposits, Nature, 346, 354-357, 1990.

Pan, H., and Symons, D.T.A., Paleomagnetism of the Mississippi Valley-type Newfoundland Zinc deposit: Evidence for Devonian or genesis and host rock remagnetization in the northern Appalachians, J. Geophys. Res., submitted.

Ravenhurst, C.E., Reynolds, P.H., Zentilli, M, and Akande, S.O., Isotope constraints on the genesis of Zn-Pb mineralization at Gays River, Nova Scotia, Canada, Econ. Geol., 82, 1294-1308, 1987.

Ravenhurst, C.E., Reynolds, P.H., Zentilli, M., Krueger, H.W., and Blenkinsop, J., Formation of Carboniferous Pb-Zn and barite mineralization from basin-derived fluids, Nova Scotia, Canada, Econ. Geol., 84, 1471-1488, 1989.

Ravenhurst, C.E., Reynolds, P.H., Zentilli, M., Krueger, H.W., and Blenkinsop, J., Strontium isotopic geochemistry of lead-zinc/barite deposits and host rocks of the Carboniferous Minas sub-basin, Nova Scotia, Geol. Sur. Can., Pap. 90-8, 245-256, 1990.

Reynolds, P.H., Elias, P., Muecke, G.K., and Grist, A.M.,Thermal history of the southwestern Meguma zone, Nova Scotia, from an $^{40}Ar/^{39}Ar$ and fission track dating study of intrusive rock, Can. J. Earth Sci., 24, 1952-1965, 1987.

Reynolds, P.H., Zentilli, M., and Muecke, G.K., K-Sr and $^{40}Ar/^{39}Ar$ geochronology of granitoid rocks from southern Nova Scotia: Its bearing on the geological evolution of the Meguma zone of the Appalachians, Can. J. Earth Sci., 18, 386-394, 1981.

Schenk, P.E., Carbonate-sulfate relations in the Windsor Group, central Nova Scotia, Canada, Internat. Cong. Carboniferous Strait. Geol.9th, Vol 3, pp. 143-162, 1984.

Scotese, C.R., Van der Voo, R., Johnson, R.E., and Giles, P.S., Paleomagnetic results from the Carboniferous of Nova Scotia, in Plate Reconstruction from Paleozoic Paleomagnetism, edited by R. Van der Voo, C.R. Scotese, and N. Bonhommet, Geodynamics Series vol. 12, AGU, Washington, D.C., 1984.

Symons, D.T.A., Bormann, R.E., and Jans, R.P., Paleomagnetism of the Triassic red beds of the lower Fundy Group and Mesozoic tectonism of the Nova Scotia platform, Canada, Tectonophys., 164, 13-24, 1989.

Van der Voo, R., Phanerozoic paleomagnetic poles from Europe and North America and comparison with continental reconstructions, Rev. Geophys., 28, 167-206, 1990.

H. Pan and D.T.A. Symons, Dept. of Geology, Univ. of Windsor, Windsor, Ontario, Canada, N9B 3P4

D.F. Sangster, Geological Survey of Canada, 601 Booth St., Ottawa, Canada, K1A 0E8

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